



Determining sample size for pilot trials: a tutorial

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Pilot trials have a key role in preparing for definitive randomized trials, yet determining their sample size remains a common challenge. This article provides practical guidance, methods, and tools—including step-by-step examples, sample size tables, and statistical code—for calculating and justifying sample sizes in external randomized pilot trials.

Definitive randomized controlled trials (known as definitive trials) are key tools for evaluating the effects of interventions. Uncompleted trials lead to research waste, delay answers to important clinical questions, and introduce bias into the medical literature.¹ About 9% of the 245 999 trials registered on ClinicalTrials.gov from 2000 to 2019 were terminated, suspended, or withdrawn.² In general, common reasons for non-completion include under-recruitment and operational problems.³⁻⁸

Unlike definitive trials, pilot and feasibility studies do not estimate or test efficacy or effectiveness, so phrases such as “preliminary efficacy” are misleading. Instead, they determine “whether something can be done, should we proceed with it, and if so, how.”⁹ Pilot and feasibility studies might or might not be randomized. For example, investigators might examine existing databases to assess enrollment feasibility by estimating the number of patients who would be eligible for a proposed definitive trial.

A pilot trial is a specific type of pilot study; it is a randomized trial in which an investigator imagines a future definitive trial and conducts a small scale version of it, either in full or in part.⁹ Pilot trials can be internal (embedded within a definitive trial, with data included in the final analysis) or external (standalone

studies, with data not contributing to the definitive trial’s analysis).¹⁰⁻¹²

If appropriately designed and interpreted, pilot trials can reduce the risk of operational failure, enhance methodological quality, and minimize bias in definitive trials.¹³ Pilot trials are part of various intervention-development frameworks, such as the ORBIT (obesity related behavioral intervention trials) model for behavioral treatment development,^{14 15} the IDEAL framework for surgical innovation,¹⁶ and the British Medical Research Council framework for complex interventions.¹⁷ In drug development, pilot trials might be conducted to evaluate the feasibility of methods planned for large phase 3 efficacy or phase 4 effectiveness trials.

Investigators are often uncertain about how many participants to include in pilot trials. Because the goal of pilot trials is not to estimate or test intervention efficacy or effectiveness, traditional sample size calculations for definitive trials are not directly applicable.¹⁸ In practice, researchers frequently rely on rules of thumb, which might be suitable for some but not all pilot trials.^{19 20} Sample size explanations are frequently absent in pilot trial reports.²¹

In this tutorial, we explain how investigators can determine and justify sample sizes for pilot trials. We focus on external pilot trials (defined above), so the methods below might not directly apply to internal pilot trials. We focus on fixed, two arm, individual randomized trial designs and do not cover more complex designs such as cluster randomized trials²²⁻²⁴ or adaptive trials.^{25 26}

We primarily describe frequentist methods because they are the most common approaches in health research. Bayesian approaches, which incorporate prior knowledge and provide probabilistic interpretations, can be especially useful for small sample sizes and post-pilot trial decision making. Such methods are increasingly being developed and discussed for pilot trial analysis and sample size calculation.²⁷⁻³⁰

This tutorial is designed to facilitate (rather than replace) conversations between investigators and biostatisticians, epidemiologists, and other key collaborators. It is organized around three possible pilot trial objectives described in the CONSORT (consolidated standards of reporting trials) extension (table 1).¹⁸ For each objective, we provide corresponding methods and examples. Sample size tables are in appendix 1. A free R Shiny application, along with the statistical code for common software (eg, R, Stata, SAS) to perform these calculations, is available online at: <https://github.com/xiying5/PilotSize>.

Inform definitive trial feasibility

Many pilot trials are designed to assess the feasibility of a future trial’s protocol by mimicking its methods.

SUMMARY POINTS

Pilot trials are conducted to assess feasibility and to inform the design of definitive trials; determining appropriate sample sizes for pilot trials can be challenging

Sample size should be determined based on a pilot trial’s objectives, and it should be aligned with the planned analysis and interpretation of the pilot trial’s results

This tutorial explains how to determine sample sizes for different pilot trial goals, and offers guidance on interpreting pilot trial results

Practical examples, sample size tables, statistical code, and an R Shiny app are included to facilitate easy implementation of the calculations

Table 1 | Overview of methods to determine pilot trial sample size

Purpose	Information needed for calculation/rule of thumb	Software implementation	Sample size table
Estimate a feasibility parameter ³¹⁻³⁴	Expected value of the parameter; desired level of precision (half-width of the confidence interval); confidence level (commonly 95%, one sided or two sided)	Precision analysis (confidence intervals for one sample): R packages such as “presize”; Stata and SAS code in GitHub	Appendix 1 table 1 (proportion); appendix 1 table 2 (Poisson rate)
Test feasibility progression criteria (traffic light framework) ³⁵	Upper and lower threshold of the progression criteria: type I error (one sided); type II error	Power analysis (one-sample test): online calculator: https://ss-progress.shinyapps.io/ss_progress_app/ (allows calculations for single-group feasibility study and pilot trials with imbalanced allocation, and evaluations of feasibility results against prespecified criteria); R packages such as EnvStats and MKpower; Stata function “power”; SPSS (power analysis, one-sample proportion); SAS function “proc power”	Appendix 1 table 3 (proportion)
		R packages such as lrrstat; Stata and SAS code in GitHub	Appendix 1 table 4 (Poisson rate)
Detect a feasibility problem ³⁶	Probability of the problem occurring in one participant; confidence level to observe the problem at least once during the pilot (95% or higher)	Online calculator (www.pilotsamplesize.com); formula 1 in appendix 2; R, Stata, and SAS code in GitHub	Appendix 1 table 5
Estimate variability	12 or 35 participants per group ^{11 37}	NA	NA
Minimize total sample size of pilot and definitive trials ^{20 38-41}	Standardized mean difference; type I error (two sided); type II error; confidence level (UCL method only; 80% or 95%)	R, Stata, and SAS code in GitHub	Appendix 1 table 6
Rule out interventions unlikely to produce clinically important effects ⁴²	Minimum clinically important difference; confidence level (one sided; commonly 80%)	Precision analysis (confidence intervals for differences between two samples): R packages such as “presize” and “statspsych”; Stata function “ciwidth”; SAS code in GitHub	Appendix 1 table 7 (mean); appendix 1 table 8 (proportion)

NA=not applicable; UCL=upper confidence limit.

They might be designed to answer questions such as the following:

- Can we achieve the required recruitment rate?
- Is randomization acceptable to the potential participants?
- Is the study team able to retain participants in the study?⁴³

Researchers use information collected in pilot trials to make informed decisions about whether and how to proceed with definitive trials.

Estimate a feasibility parameter

Judgments about feasibility often depend on one or more specific parameters (eg, recruitment rate, dropout, willingness to be randomized).⁹ Because these parameters are often unknown, researchers conduct pilot trials to estimate them. To determine the sample size needed for a precise estimate, three values must be specified as in worked examples (box 1)³¹⁻³⁴:

- The expected value of the parameter (eg, 10% of participants will drop out of the trial)
- The desired level of precision or half-width of the confidence interval (CI; eg, $\pm 10\%$) around the parameter
- The confidence level (commonly 95%, either one sided or two sided, depending on whether estimates in both directions are of interest).

The expected value (ie, step 1) often requires an educated guess. The desired level of precision (ie, step 2) might depend on the importance of a parameter. For example, investigators in one setting might be confident in their ability to recruit participants based on past experience, but they might want to estimate dropout very precisely because it could have a large

impact on the definitive trial’s sample size.³² In another setting, investigators might be more concerned about recruitment compared with dropout.

Estimating these parameters can provide valuable information about the feasibility of a planned definitive trial. For instance, using a 95% confidence level and ± 10 percentage points precision, researchers would expect that the true value of a parameter is sufficiently likely to be within ± 10 percentage points of the pilot estimate. This approach assumes that the pilot and definitive trials share the same design, methods, procedures, staff, population, sites, and so on.

Estimating certain feasibility parameters with precision (eg, recruitment rate, dropout) might also inform the definitive trial’s sample size calculation. However, this approach might not directly inform decisions on whether to proceed to a definitive trial or not.

Test feasibility progression criteria

CONSORT guidelines recommend that researchers prespecify criteria to determine if progression from a pilot trial to future a definitive trial is warranted.¹⁸ Use of progression criteria that are directly tied to a formative plan⁴⁴ for a future randomized controlled trial increases the chances that the definitive trial will be conducted successfully.⁴⁵ Mellor and colleagues propose a set of recommendations for progression criteria,⁴⁶ whereby researchers can use traffic light set of progression criteria to follow three steps^{12 18 35 47}:

1. Identify one or more feasibility parameters, which are core markers of whether a definitive trial could successfully answer its primary research question.⁴⁸
2. Set a minimum threshold and a goal threshold for each parameter.

Box 1: Estimate a feasibility parameter (worked examples)**Example 1: Dropout probability**

To understand the extent to which participants might drop out of the definitive trial, researchers design a pilot trial to estimate the probability of dropout.

- They make an educated guess that dropout will be about 10%.
- They aim to estimate this parameter with a precision of 10 percentage points (ie, confidence interval (CI) width is 20 percentage points).
- They plan to use a one sided 95% CI because they are mostly interested in the upper bound of the CI (ie, whether it exceeds 20%).

The researchers will need to enroll 26 participants (13 per group) for the pilot trial to have a 95% CI with a maximum width of 20% (0% to 20%; appendix 1 table 1).

After the pilot trial, researchers calculate the point estimate of the proportion who drop out and its one sided 95% CI. This CI can be interpreted as a range of probable values for the true dropout probability, given the study design, method, setting, and everything else.

Example 2: Recruitment rate

Researchers conduct a traditional sample size calculation for the definitive trial and find that they would need to enroll 180 participants over 18 months (ie, 10 enrollments per month). Because participant recruitment is crucial for trial success, they plan to conduct a pilot trial to estimate the recruitment rate.

- The recruitment rate needed for successful enrollment in the definitive trial is 10 participants per month.
- They aim to estimate this parameter with a precision of 3 (ie, CI width is six participants per month).
- They plan to use a two sided 95% CI.

To estimate this recruitment rate with the desired level of precision, researchers calculate that 44 participants (22 per group) will need to be enrolled over 4.4 months during the pilot trial (appendix 1 table 2). This calculation assumes a constant recruitment rate, and researchers might choose to extend the duration of the pilot trial if they believe that this is not enough time to initiate and stabilize recruitment. The pilot trial can be stopped when either the target number of participants is reached or when the planned enrollment period is met. Alternatively, the pilot trial may continue until both criteria are fulfilled to gather additional data. Following the pilot trial, the researchers will calculate a point estimate of the recruitment rate along with its two sided 95% CI.

3. Interpret the results as:

- Go (green): if the results exceed the goal threshold for all parameters (compare the point estimate to the threshold; P value not needed), proceed to the definitive trial with the original design
- Amend (yellow): if between the minimum and goal thresholds for one or more parameters (but not below the minimum threshold for any parameter), modify aspects of the methods or design to improve feasibility, or more specifically:
 - Minor amend: if $P \text{ value} < \alpha$ (commonly 0.05)
 - Major amend: if $P \text{ value} \geq \alpha$ (commonly 0.05)
- Stop (red): if below the minimum for one or more parameters (compare the point estimate to the threshold; P value not needed), do not proceed.

Box 2 shows some worked examples of these criteria.

Example 2: Recruitment rate

Investigators anticipate that a definitive trial will require 240 participants and have 24 months to recruit. To assess recruitment feasibility, they plan a pilot trial. Their goal threshold is 10 enrollments per month, and their minimum threshold is six per month. If recruitment falls between six and 10, they expect that they will be able to adjust strategies to reach 10 per month and/or to extend the duration of the trial to achieve the required sample size. However, if it drops

below six, they anticipate difficulty enrolling 240 participants within the available time.

The researchers decide to use the three tier progression criteria for quick decision making. Because the unit of observation is month rather than individuals, they calculate that an enrollment period of five months is needed for the pilot trial. This calculation is based on a one-sample exact test for Poisson rate with a one sided type I error of 0.05 and power of 0.9 (appendix 1 table 4). It assumes a constant recruitment rate, so researchers might choose to extend the duration of the pilot trial to allow recruitment to stabilize.

On completing the pilot trial, researchers will compare observed recruitment against a null hypothesis of six participants per month, and decide the next steps as follows:

- Go: if the point estimate of recruitment rate is greater than 10 participants per month, the P value will be below 0.05 and they will proceed with the definitive trial.
- Amend: If the point estimate is between six and 10 participants per month, they will modify the recruitment strategy, regardless of the P value.
- Stop: If the point estimate is below six participants per month, the P value will exceed 0.05 and they will halt plans for the definitive trial.

For both the three tier (go, amend, stop) and four tier (go, minor amend, major amend, stop) progression criteria, Lewis and colleagues³⁵ suggest determining the pilot sample size for each feasibility parameter using a one sided test of the parameter's null value

Box 2: Test feasibility progression criteria (worked examples)**Example 1: Recruitment uptake**

Researchers are uncertain about whether eligible participants will consent to participate. They plan to conduct a pilot trial and set the following goal and minimum thresholds:

- Goal threshold: if 50% of eligible participants consent to participation, the definitive trial is highly likely to succeed in recruitment
- Minimum threshold: if consent is below 20%, a definitive trial is unlikely to succeed.

The researchers decide to use the four tier progression criteria for nuanced decision making. Using a one-sample proportion test with a one sided type I error of 0.05 and power of 0.95, researchers calculate a required pilot sample size of 28 participants (14 per group; appendix 1 table 3).

On completion of the pilot trial, researchers will test the observed probability of consent in the pilot trial against the null hypothesis of 20%. They will make the decision about whether and how to progress to a definitive trial as follows:

- Go: if the point estimate surpasses 50%, the associated P value will be below 0.05 and the definitive trial will be conducted.
- Minor amend: if the point estimate falls between 20% and 50%, and the one sided P value is below 0.05, the researchers will modify the recruitment strategy with minor changes.
- Major amend: if the point estimate falls between 20% and 50%, and the one sided P value is above 0.05, the researchers will modify the recruitment strategy with major changes.
- Stop: if the point estimate is below 20%, the associated P value will exceed 0.05 and the researchers will not conduct the definitive trial.

(the minimum threshold) versus the alternative (goal threshold) with at least 80% power and one sided α of 0.05. The four tier approach uses the statistical significance threshold in the appraisal and hence fully links the sample size calculation with the evaluation.

If sample size calculations are done for multiple parameters, power should be adjusted to account for type II error inflation and the largest sample size should be used. For example, the probability of meeting two criteria powered at 80% is only 64% (ie, 0.8^2), so an appropriate strategy would be to power each criterion at 90% to achieve collective power of 80%.

Detect a feasibility problem

Some pilot trials aim to detect events that could compromise the quality or flow of the definitive trial, such as logistical problems that disrupt trial workflows and technical failures with data collection procedures.⁴⁹ Viechtbauer and colleagues propose determining the number of participants needed for a pilot trial to achieve a desired confidence level of observing one or more problematic events.³⁶ This calculation is based on an assumed minimum probability of such events could jeopardize the success of the definitive trial and thus warrant detection (appendix 2 formula 1).³⁶ This approach can also be applied to phase 2 trials that aim to identify uncommon adverse events.⁵⁰ As in a worked example (box 3), researchers must specify:

- The minimum probability of the event occurring in an individual participant that would be problematic; and

- The desired level of confidence or certainty in detecting at least one event. Typically, a high confidence level (eg, 95%) is used.

This method is best suited for issues that could jeopardize definitive trials even if they occur with low but non-negligible probabilities (eg, 1% to 10%).³⁶ Rare events (eg, <1 in 1000) might require impractically large pilot samples to have a reasonable chance of detection. Notably, the aim is not to estimate the event probability. If no events occur in the pilot trial, researchers can only conclude that the true probability might be lower than expected, but the exact number remains undetermined.

Inform definitive trial sample size

When planning a definitive trial, researchers typically need to determine an appropriate sample size to ensure the study has adequate statistical power. This process, known as power analysis, requires information such as effect size, variability, significance level, desired power, and feasibility (eg, recruitment rate, proportion of individuals who drop out). While pilot trials can provide useful data for some parameters (eg, variability, feasibility), pilot trials are not suitable for determining effect sizes. By definition, pilot trials have small sample sizes that result in imprecise estimates of effects, and pilot trials with promising results might be more likely to be published and cited in plans for definitive trials, so relying solely on effect point estimates from pilot trials can lead to suboptimal sample sizes for definitive trials.⁵¹⁻⁵⁴ Rather, researchers should use clinically important differences between groups—informed by previous research and expert opinion—to determine effect sizes that should be used to calculate required sample sizes.

Estimate variability

Variability describes how much individual values in a group differ from the average value (ie, standard deviation or variance). Variability is often regarded as a nuisance parameter that is not of direct interest but must be accounted for in the analysis. Underestimating variability of the primary outcome can lead to an underpowered definitive trial that is too small to identify real effects of clinical importance. Overestimating variability can lead to an unnecessarily large trial, wasting resources. In general, variance estimates might come from studies with similar designs, populations, and analyses to the planned trial.⁵⁵ If possible, researchers might combine data from multiple studies (eg, using meta-analysis) to estimate variance. Pilot trials can estimate variability, but their small sizes make estimates imprecise. Relying on such estimates can result in definitive trials that are either underpowered or overpowered. Researchers are typically more concerned about underpowering than overpowering definitive trials.

To estimate variability for normally distributed outcomes, researchers can conduct a pilot trial with 12 or 35 participants per group to achieve reasonable precision and efficiency (appendix 2 box 1 includes

Box 3: Detect a feasibility problem (worked example)

Researchers plan to conduct a definitive trial with a primary outcome based on the presence or absence of a specific condition. Diagnoses of this condition will be made independently by two clinicians. If they disagree, additional assessments, which are more costly and time consuming, will be required. The researchers are concerned about the likelihood of disagreement and aim to study this probability.

- The researchers assume that if the clinicians disagree on 10% or more of the diagnoses, the expected cost and duration of the definitive trial will be exceeded by a considerable amount.
- They plan to use a 95% confidence level.

To ensure 95% confidence in observing at least one disagreement, the researchers calculate a sample size of 30 participants (15 per group for a two-group study, rounded up from 29 for 1:1 equal allocation) for the pilot trial (appendix 1 table 5).

The results of the pilot trial will be interpreted as follows:

- No disagreements: If no disagreements occur during the pilot, the actual probability of disagreement is likely lower than 10%, making it unlikely that the projected costs and study duration will be exceeded considerably.
- One or more disagreements: If one or more disagreements are observed, the actual probability of disagreement could be 10% or higher, presenting a potential challenge for the definitive trial.

more details).^{11 37} Conservatively, the definitive trial's sample size should be increased to mitigate the probability of being underpowered because the variability estimate comes from a small trial.³⁸ When there is uncertainty in the variance estimate, blinded sample size re-estimation might be done after the definitive trial has started⁵⁵; this approach uses the variance data collected during the trial to adjust the sample size as needed.⁵⁶

Inflation of the definitive trial sample size can be done in one of two approaches (appendix 2 table 1 includes simulation details):

- Use 90% power in the sample size calculation for the definitive trial to achieve 80% power most of the time—over 90% with 35 individuals per group (preferable when feasible) and about 80% with 12 individuals per group (acceptable if resources are limited).

Box 4: Minimize the total sample size of pilot and definitive trials (worked example)

Researchers plan to test an intervention with a continuous primary outcome in a definitive trial and a standardized mean difference of 0.4, which is the minimum effect of clinical importance. To calculate the pilot sample size that minimizes the overall sample size:

1. The researchers start with a small pilot of 10 per group.
2. They calculate that the definitive trial needs 100 participants per group, assuming a standardized mean difference of 0.4, a two sided type I error rate of 0.05, and a type II error rate of 0.2. They then inflate this size to 139 per group using the 80% confidence limit method.³⁹
3. The combined sample size of both the pilot and definitive trials is 149 participants per arm.
4. They increase the pilot size by one participant per group and repeat the above three steps.
5. The researchers find that to minimize the combined sample size, the pilot trial should enroll 36 participants (18 per group; appendix 1 table 6 includes sample sizes based on other inflation methods).

- Adjust the sample size for the definitive trial upward by using the 80% or 95% upper confidence limit of the variance estimate from the pilot trial to account for uncertainty,^{20 38-40} or acknowledge that the variance estimate is based on a sample (ie, not the population) in the calculation.^{40 41} Appendix 2 box 2 includes details on these inflation methods.

Minimize the total sample size of pilot and definitive trials

Because pilot trials are part of larger research programs, researchers might be interested in minimizing the combined sample sizes of pilot and subsequent definitive trials. This approach can be achieved by calculating the definitive trial sample size based on the variability observed in the pilot trial and adjusting it using the inflation methods mentioned above.^{20 38-41} The trade-off between information from the pilot trial and the definitive trial produces a U shaped curve³⁸ for the overall sample size requirement. Specifically, as the pilot trial size increases, the adjusted size of the definitive trial decreases, which reduces the total sample size up to a point. Beyond that point, further increases in the pilot size outweigh the decreases in the adjusted definitive trial size. Thus, a pilot trial size exists that minimizes the total number of participants needed for both trials. Deriving the pilot sample size that minimizes the total sample size requires an iterative process^{20 38-41}:

1. Start with a small pilot sample size (eg, eight individuals per group).
2. Use standard methods to calculate the definitive trial's sample size and inflate it using one of the inflation methods (appendix 2 box 2).^{20 38-41}
3. Calculate the total sample across pilot and definitive trial.
4. Increase pilot size and repeat steps 2-3.
5. Identify pilot size that minimizes the overall total sample size.

To use this approach, researchers must first define an effect of clinical interest, typically through literature review and expert knowledge. Researchers also need to specify variance estimates, which might be derived from previous similarly designed trials. If such estimates are unavailable, a standardized mean difference between groups can be used in step 2. If a specific standardized difference cannot be determined, stepped rules can be applied: 75, 25, 15, and 10 participants per group for standardized mean differences that are very small (≤ 0.1), small (0.2), medium (0.5), or large (0.8), respectively,^{40 57} or a total pilot sample size of 55 for small to medium effect sizes.³⁸ Box 4 includes a worked example.

Rule out interventions unlikely to produce clinically important effects

Researchers and funders often want to rule out interventions that are unlikely to have clinically important effects before launching definitive trials.

Box 5: Rule out interventions unlikely to produce clinically important effects (worked examples)**Continuous outcome**

Researchers plan to assess a weight loss program in a definitive trial. They first want to conduct a pilot trial to evaluate the feasibility of the trial protocol. The investigators identify a minimum clinically important effect based on literature and expert opinions. Because the intervention is novel, and because a definitive trial would be expensive, they also want to evaluate whether the intervention is unlikely to produce the minimum clinically important effect. To calculate the sample size, they follow these steps:

1. They assume no effect of the intervention, setting a mean difference of zero between the weight loss group and control group.
2. They decide on a standardized effect size of 0.3 for the primary outcome. Based on previous studies and expert opinion, this is the smallest effect size they consider meaningful for a definitive trial.
3. They choose to use an 80% one sided confidence interval (CI) for their calculations.

The researchers calculate that the pilot trial needs 32 participants (16 per group) to yield a one sided 80% CI of the effect size with an upper limit below 0.3 when centered around zero (appendix 1 table 7). If the effect estimate in the pilot trial is greater than zero, and the trial protocol is feasible according to other criteria, they might proceed to the subsequent trial.

Binary outcome

Researchers aim to assess a vaccine uptake intervention and want to exclude it from further testing if it is unlikely to have a minimum clinically important effect:

1. They assume an equal proportion of vaccinated people in both groups, starting at 50%.
2. They expect the intervention to increase vaccination rates from 50% to 60% (or more).
3. They use this 10% difference as the maximum half width of a one sided 80% CI.

Based on these parameters, the researchers calculate that the pilot trial needs to include 70 participants (appendix 1 table 8). This sample size ensures that the half width of the one sided 80% CI would be less than 10%. If the effect estimate in the pilot trial is greater than zero, and the trial protocol is feasible (and so on), the researchers might proceed to the subsequent trial.

This approach can help avoid waste and minimize exposure of participants to interventions that could be ineffective or harmful. For health conditions unlikely to improve without treatment, single group proof-of-concept studies might be used to assess whether an intervention is plausible to achieve clinically significant results and warrant further investigation. In these studies, researchers establish targets for meaningful improvement and calculate the proportion of participants who meet those targets.¹⁴ In some contexts, researchers might conduct pilot trials of appropriate size to inform decisions about progression.⁴² This approach relies on the sample size calculation method proposed by Cocks and Torgerson⁴² that accounts for additional uncertainty in the effect size estimate. It does not provide a precise estimate or allow for rigorous hypothesis testing, so researchers cannot determine whether an intervention is effective based on pilot trials alone. When researchers aim to estimate an effect precisely or to conduct hypothesis tests, but randomizing a large sample is not feasible, researchers should consider other study designs such

as fully powered phase 2 efficacy trials using surrogate outcomes, bayesian designs that incorporate prior data or knowledge, or rigorous non-randomized studies.

For pilot trials with continuous outcomes, researchers would:

1. Assume a mean difference of zero between the treatment and control groups.
2. Decide on a minimum clinically important standardized difference between groups (eg, standardized mean difference of 0.3).
3. Set the minimum clinically important difference as the maximum half width of the 80% or 90% one sided CI around the zero effect (one sided CI is adequate because only one direction is of interest).

For binary outcomes, researchers would:

1. Assume an equal proportion of participants experiencing the outcome in both groups (eg, 30%). If the probable proportions are not known, a proportion of 50% could be used to maximize the sample size.⁵⁸
2. Decide on a minimum clinically important difference between groups (eg, 10 percentage points).
3. Use the minimum clinically important difference as the maximum half width of the 80% or 90% one sided CI.

Then, the pilot trial's sample size would be calculated so that it is sufficiently large to confidently rule out a clinically important effect if the intervention has no effect or a harmful effect. After the pilot trial, no hypothesis testing would be conducted. Researchers would simply compare the point estimate of the between-group difference to zero. If the observed difference were larger than zero—and if pilot trial results suggest that the trial protocol is feasible, and so on—then the researchers would conclude that they cannot confidently rule out the possibility that the intervention has a clinically significant benefit relative to the comparator, and they might then progress to a definitive trial. If the point estimate were zero or below, the researchers might not proceed. Box 5 shows worked examples for pilot trials with continuous or binary outcomes.

This approach maintains a low probability of observing a negative point estimate unless the true effect is substantially below clinical relevance, potentially ruling out ineffective interventions and informing progression decisions. However, there is a risk of mistakenly discarding beneficial interventions. For example, using 80% CIs, interventions with the expected clinically important effects will be discarded 20% of the time. Conversely, positive point estimates do not confirm that true effect sizes are clinically important or even positive; true effects could be smaller than, equal to, or larger than thresholds. Therefore, while some investigators might consider that the level of precision provided by a definitive trial is not necessary for making

go or no-go decisions, others view this approach as insufficient to definitively rule out clinically relevant effects. A no-go decision from a pilot trial might not be interpreted as a definitive end to an intervention's development and testing.⁴² Instead, researchers might use the findings to guide further adjustments.⁴²

Discussion

This tutorial provides general guidance for determining sample sizes in pilot trials. It should help researchers understand different approaches to designing pilot trials and to have informed conversations with biostatisticians, epidemiologists, and other methodologists when designing pilot trials to answer specific research questions.

This tutorial does not relate to other pilot and feasibility studies that often complement pilot trials. Moreover, it does not provide an exhaustive account of analytical methods that researchers might use in pilot trials to inform the definitive trials. For example, when assessing recruitment rates, researchers might calculate the monthly recruitment rate and compare it to the target rate to evaluate how often it meets or exceeds expectations. Meeting a target three out of three times might be more convincing than three out of four times. However, recruitment rates can vary over time—starting slowly, peaking, and then declining as interest wanes—so results should be interpreted with caution.

Sample size calculations for pilot trials should align with study objectives, and a single pilot trial might have multiple goals. In such cases, multiple sample sizes might be calculated and the largest sample size might be selected. Ultimately, the decision to proceed to a definitive trial should be based on a holistic evaluation of all aspects such as feasibility, available resources, and scientific importance, rather than relying solely on statistical criteria.⁵⁹

Pilot trial results might not always generalize fully to definitive trials. Even if eligibility criteria are the same, participants enrolled in a pilot and definitive trial might differ if neither includes a large, representative sample of the target population. Additionally, researchers often modify trial methods or interventions based on pilot trial findings, which can limit the applicability of pilot trial results.⁶⁰ While conducting a second pilot trial to test the effects of changes is uncommon, researchers should identify and explain how modifications might affect definitive trials.⁴⁵

Identifying the reasons for conducting pilot trials, and following the appropriate steps to determine sample sizes, might help investigators collaborate with statisticians and methodologists, design more informative pilot trials, justify their decisions to funders and ethics committees, better interpret pilot trial results, and conduct more successful definitive trials.

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Web appendix 1: Sample size tables

Web appendix 2: Supplementary materials